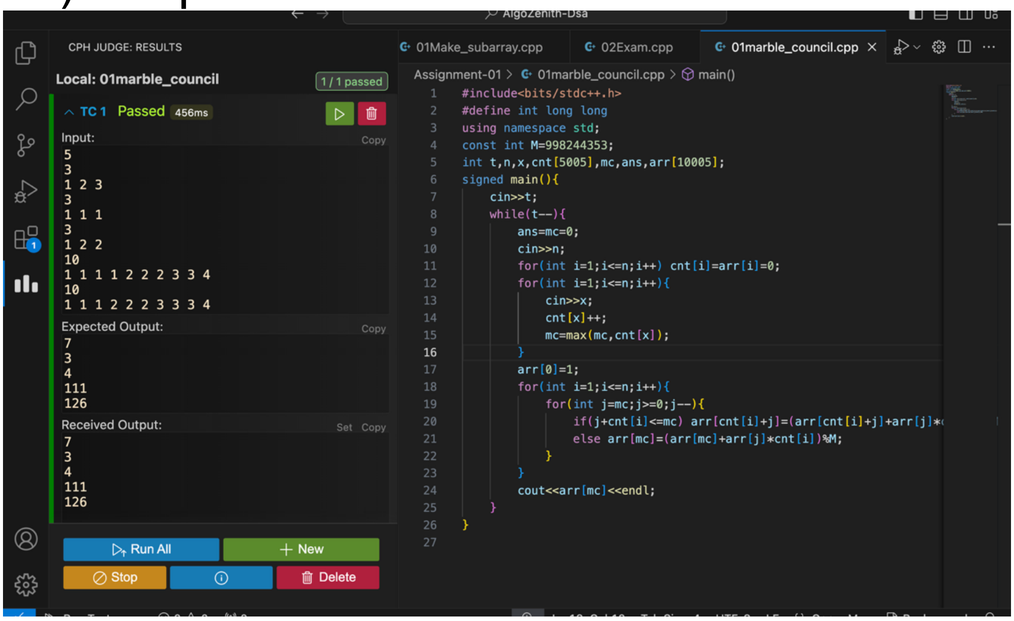
Competitive Coding-3 Assignments

Assignment – 01

01) Code:

```
#include<bits/stdc++.h>
#define int long long
using namespace std;
const int M=998244353;
int t,n,x,cnt[5005],mc,ans,arr[10005];
signed main(){
    cin>>t;
    while(t--){
        ans=mc=0;
        cin>>n;
        for(int i=1;i<=n;i++) cnt[i]=arr[i]=0;
        for(int i=1;i<=n;i++){
            cin>>x;
            cnt[x]++;
            mc=max(mc,cnt[x]);
        }
        arr[0]=1;
        for(int i=1;i<=n;i++){
            for(int j=mc;j>=0;j--){
                if(j+cnt[i]<=mc) arr[cnt[i]+j]=(arr[cnt[i]+j]+arr[j]*cnt[i])%M;
                else arr[mc]=(arr[mc]+arr[j]*cnt[i])%M;
            }
        }
        cout<<arr[mc]<<endl;
    }
}
```

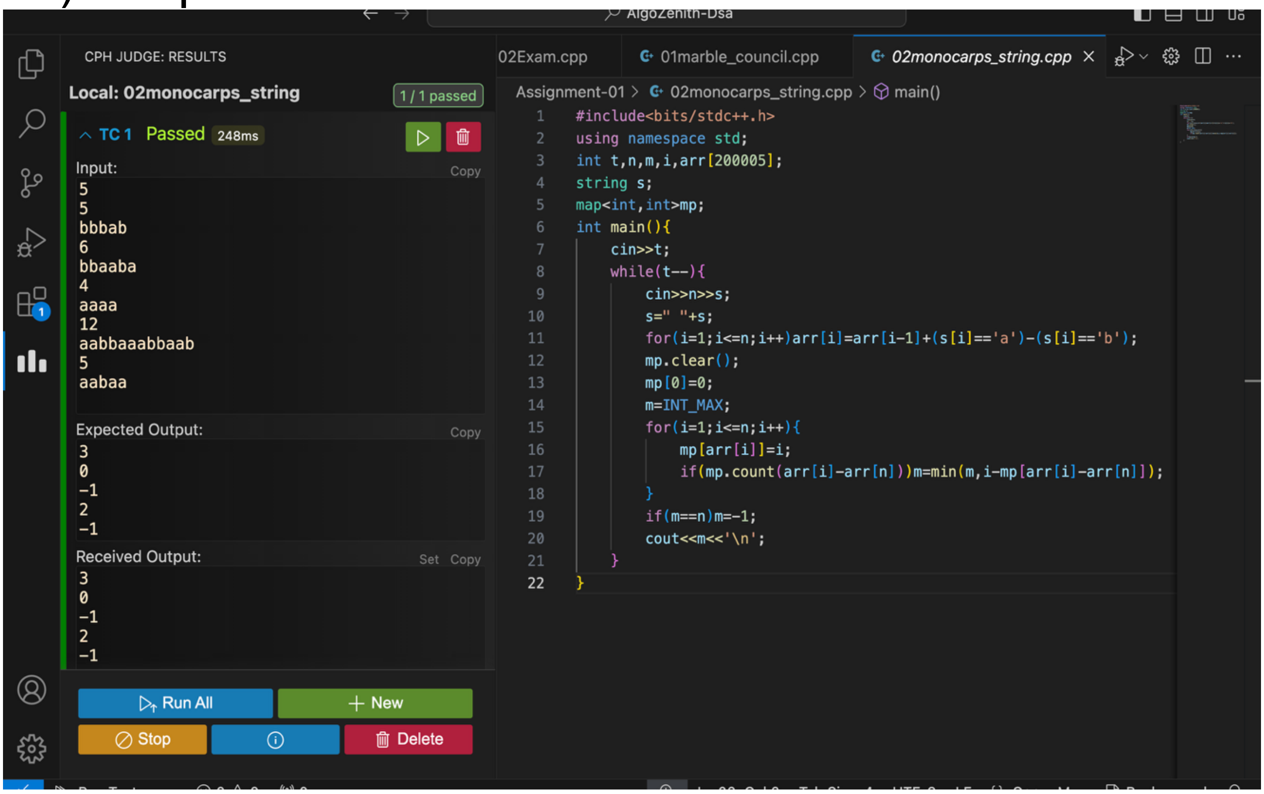
01) Output:



02) Code:

```cpp
#include<bits/stdc++.h>
using namespace std;
int t,n,m,i,arr[200005];
string s;
map<int,int>mp;
int main(){
    cin>>t;
    while(t--){
        cin>>n>>s;
        s=" "+s;
        for(i=1;i<=n;i++)arr[i]=arr[i-1]+(s[i]=='a')-(s[i]=='b');
        mp.clear();
        mp[0]=0;
        m=INT_MAX;
        for(i=1;i<=n;i++){
            mp[arr[i]]=i;
            if(mp.count(arr[i]-arr[n]))m=min(m,i-mp[arr[i]-arr[n]]);
        }
        if(m==n)m=-1;
        cout<<m<<'\n';
    }
}
```

02) Output:

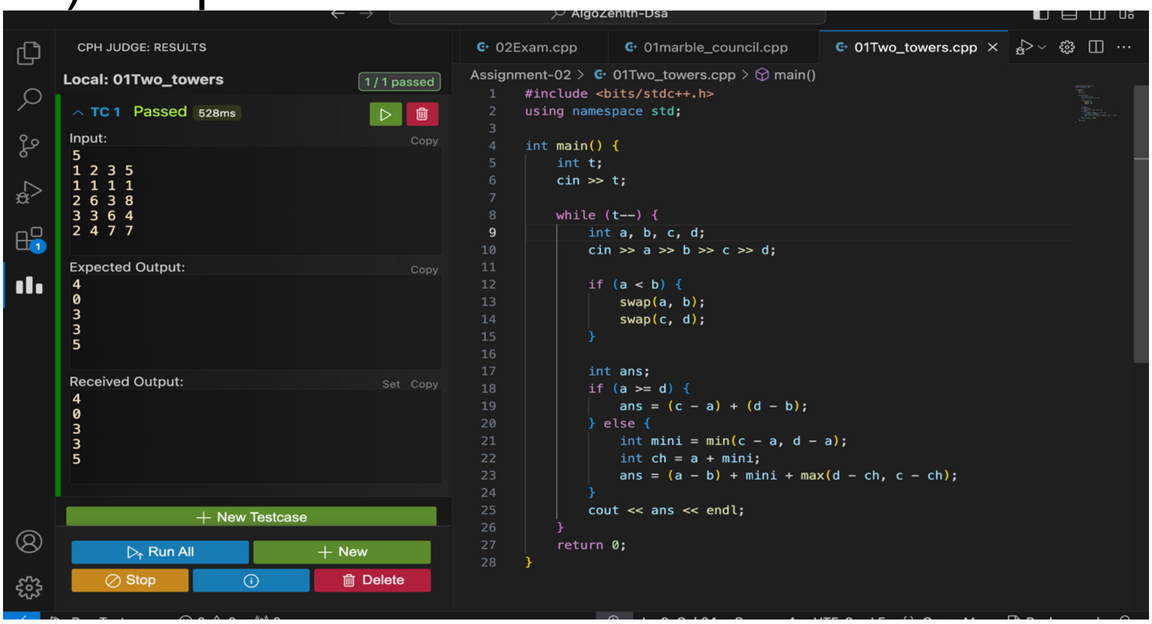


Assignment – 02

01) Code:

```cpp
#include <bits/stdc++.h>
using namespace std;

int main() {
    int t;
    cin >> t;

    while (t--) {
        int a, b, c, d;
        cin >> a >> b >> c >> d;

        if (a < b) {
            swap(a, b);
            swap(c, d);
        }

        int ans;
        if (a >= d) {
            ans = (c - a) + (d - b);
        } else {
            int mini = min(c - a, d - a);
            int ch = a + mini;
            ans = (a - b) + mini + max(d - ch, c - ch);
        }
        cout << ans << endl;
    }
    return 0;
}
```

01) Output:

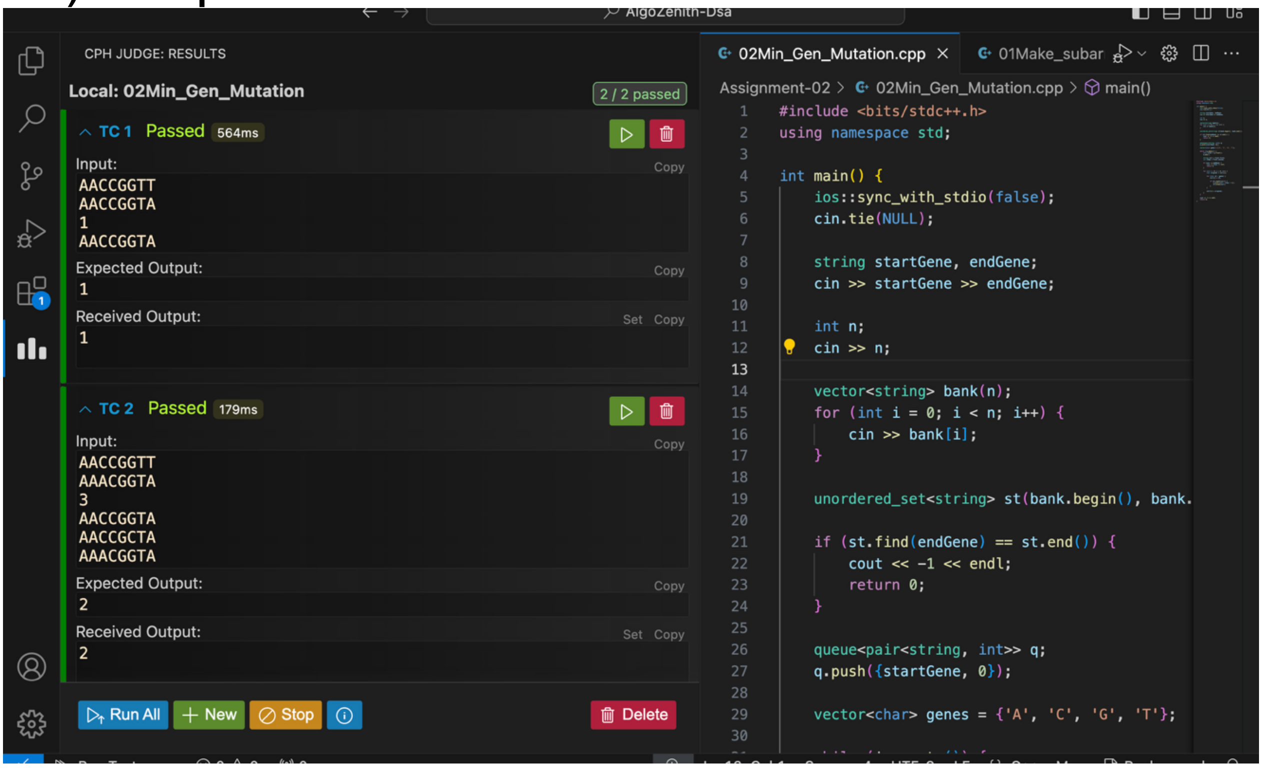


02)Code:

```
#include <bits/stdc++.h>
using namespace std;

int main() {
    ios::sync_with_stdio(false);
    cin.tie(NULL);

    string startGene, endGene;
    cin >> startGene >> endGene;

    int n;
    cin >> n;

    vector<string> bank(n);
    for (int i = 0; i < n; i++) {
        cin >> bank[i];
    }

    unordered_set<string> st(bank.begin(), bank.end());

    if (st.find(endGene) == st.end()) {
        cout << -1 << endl;
        return 0;
    }

    queue<pair<string, int>> q;
    q.push({startGene, 0});

    vector<char> genes = {'A', 'C', 'G', 'T'};

    while (!q.empty()) {
        auto front = q.front();
        q.pop();

        string curr = front.first;
        int steps = front.second;

        if (curr == endGene) {
            cout << steps << endl;
            return 0;
        }

        for (int i = 0; i < 8; i++) {
            char original = curr[i];

            for (char ch : genes) {
                curr[i] = ch;

                if (st.count(curr)) {
```

```
                    q.push({curr, steps + 1});
                    st.erase(curr);
                }
            }

            curr[i] = original;
        }
    }

    cout << -1 << endl;
    return 0;
}
```

02) Output:

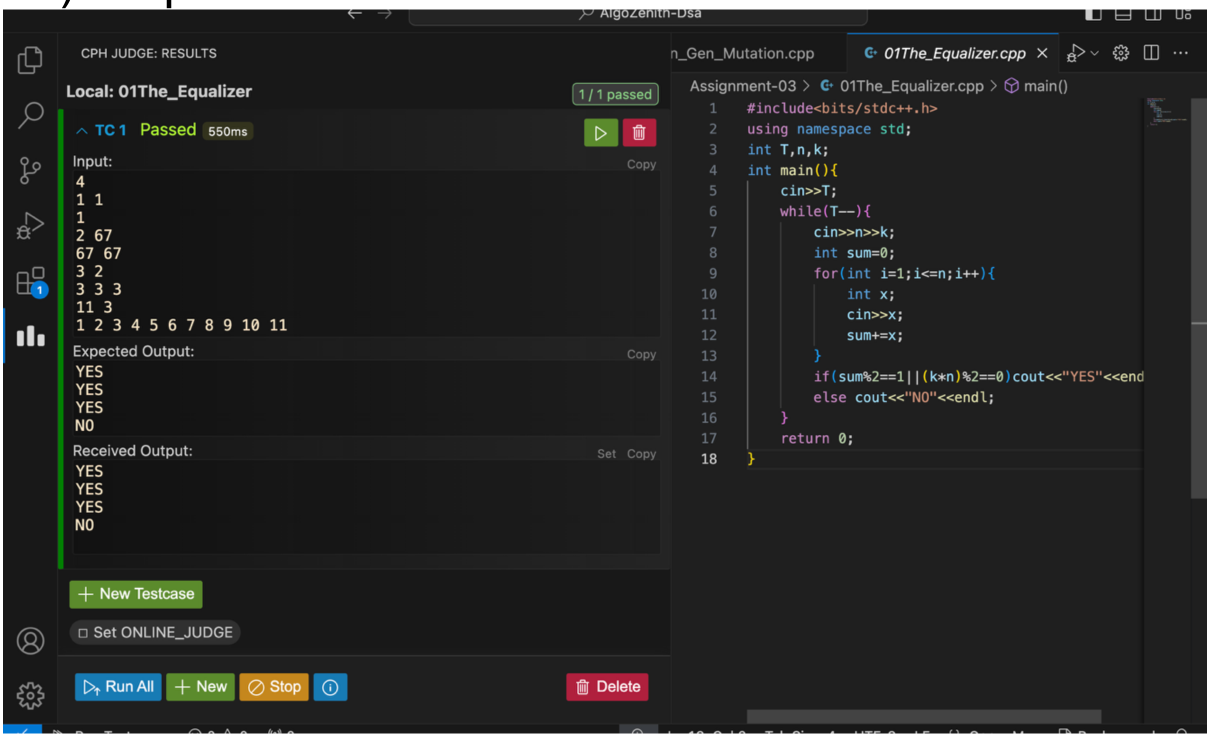


Assignment – 03

01) Code:

```cpp
#include<bits/stdc++.h>
using namespace std;
int T,n,k;
int main(){
    cin>>T;
    while(T--){
        cin>>n>>k;
        int sum=0;
        for(int i=1;i<=n;i++){
            int x;
            cin>>x;
            sum+=x;
        }
        if(sum%2==1||(k*n)%2==0)cout<<"YES"<<endl;
        else cout<<"NO"<<endl;
    }
    return 0;
}
```

01)Output:



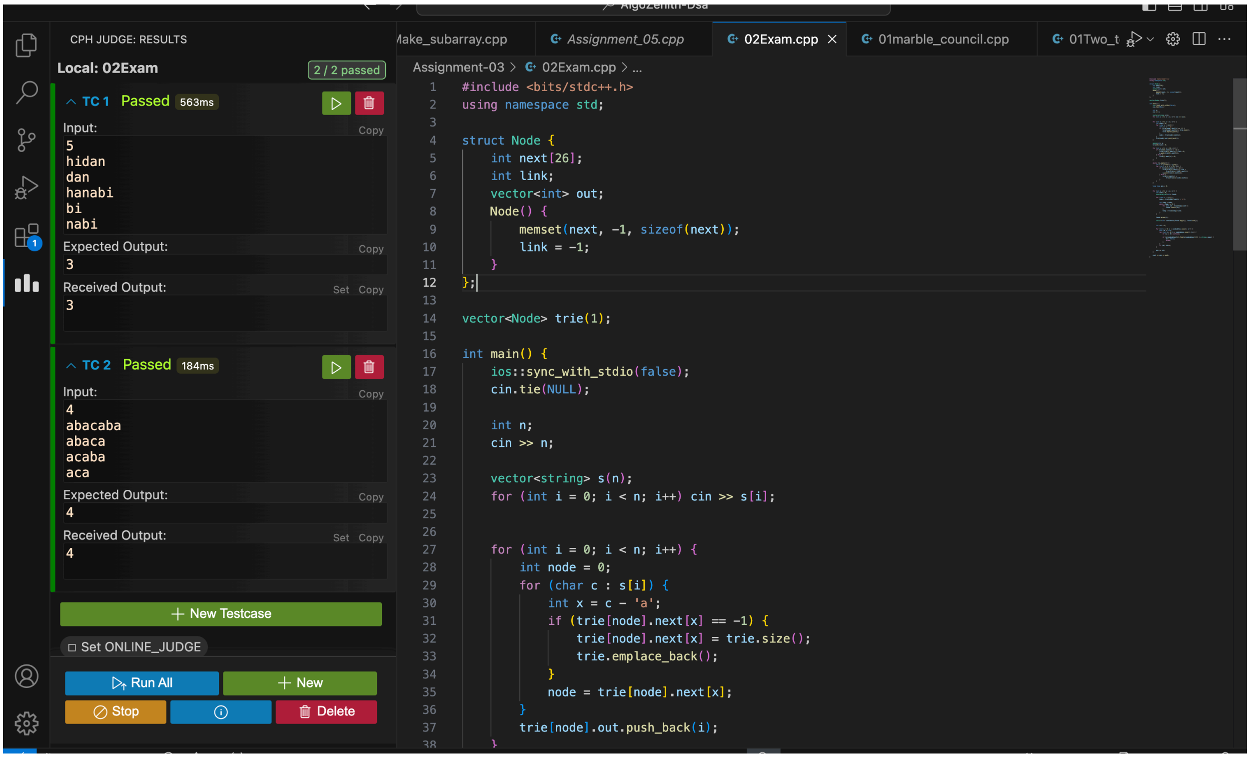
02) Code:

```
#include <bits/stdc++.h>
using namespace std;

struct Node {
    int next[26];
    int link;
    vector<int> out;
    Node() {
        memset(next, -1, sizeof(next));
        link = -1;
    }
};

vector<Node> trie(1);

int main() {
    ios::sync_with_stdio(false);
    cin.tie(NULL);

    int n;
    cin >> n;

    vector<string> s(n);
    for (int i = 0; i < n; i++) cin >> s[i];


    for (int i = 0; i < n; i++) {
        int node = 0;
        for (char c : s[i]) {
            int x = c - 'a';
            if (trie[node].next[x] == -1) {
                trie[node].next[x] = trie.size();
                trie.emplace_back();
            }
            node = trie[node].next[x];
        }
        trie[node].out.push_back(i);
    }

    queue<int> q;
    trie[0].link = 0;

    for (int c = 0; c < 26; c++) {
        if (trie[0].next[c] != -1) {
            trie[trie[0].next[c]].link = 0;
            q.push(trie[0].next[c]);
        } else {
            trie[0].next[c] = 0;
        }
```

```
    }

    while (!q.empty()) {
        int v = q.front(); q.pop();
        for (int c = 0; c < 26; c++) {
            if (trie[v].next[c] != -1) {
                trie[trie[v].next[c]].link =
                    trie[trie[v].link].next[c];
                q.push(trie[v].next[c]);
            } else {
                trie[v].next[c] =
                    trie[trie[v].link].next[c];
            }
        }
    }

    long long ans = 0;


    for (int i = 0; i < n; i++) {
        int node = 0;
        unordered_set<int> found;

        for (char c : s[i]) {
            node = trie[node].next[c - 'a'];

            int temp = node;
            while (temp != 0) {
                for (int id : trie[temp].out) {
                    found.insert(id);
                }
                temp = trie[temp].link;
            }
        }

        found.erase(i);

        vector<int> candidates(found.begin(), found.end());


        int cnt = 0;

        for (int j = 0; j < candidates.size(); j++) {
            bool ok = true;
            for (int k = 0; k < candidates.size(); k++) {
                if (j == k) continue;

                if (s[candidates[k]].find(s[candidates[j]]) != string::npos) {
                    ok = false;
                    break;
                }
```

```
            }
            if (ok) cnt++;
        }

        ans += cnt;
    }

    cout << ans << endl;
}
```

02) Output:

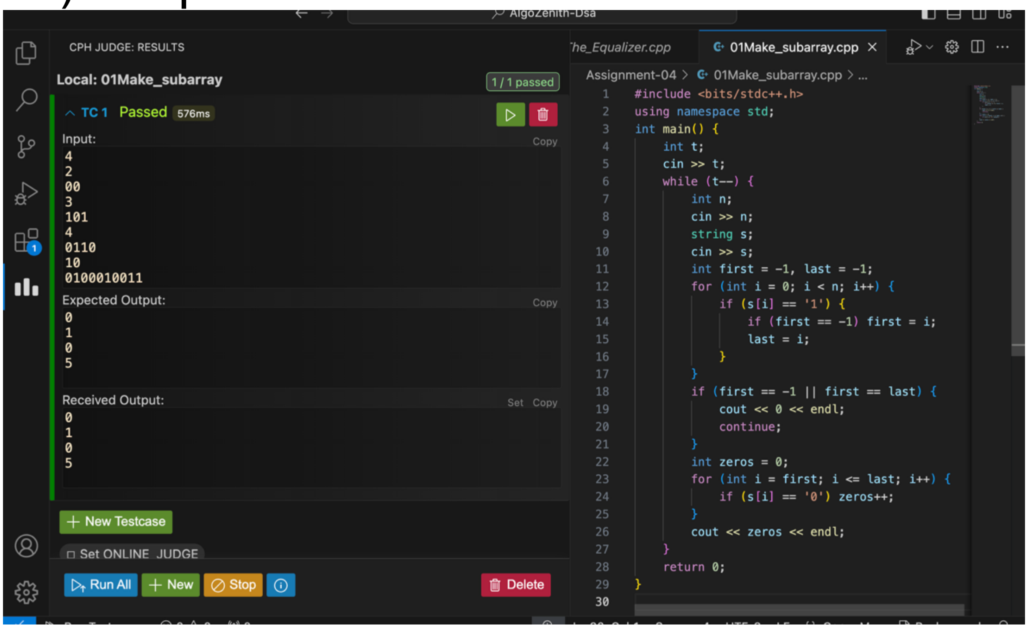


Assignment-04

01) Code:

```cpp
#include <bits/stdc++.h>
using namespace std;
int main() {
    int t;
    cin >> t;
    while (t--) {
        int n;
        cin >> n;
        string s;
        cin >> s;
        int first = -1, last = -1;
        for (int i = 0; i < n; i++) {
            if (s[i] == '1') {
                if (first == -1) first = i;
                last = i;
            }
        }
        if (first == -1 || first == last) {
            cout << 0 << endl;
            continue;
        }
        int zeros = 0;
        for (int i = first; i <= last; i++) {
            if (s[i] == '0') zeros++;
        }
        cout << zeros << endl;
    }
    return 0;
}
```

01) Output:

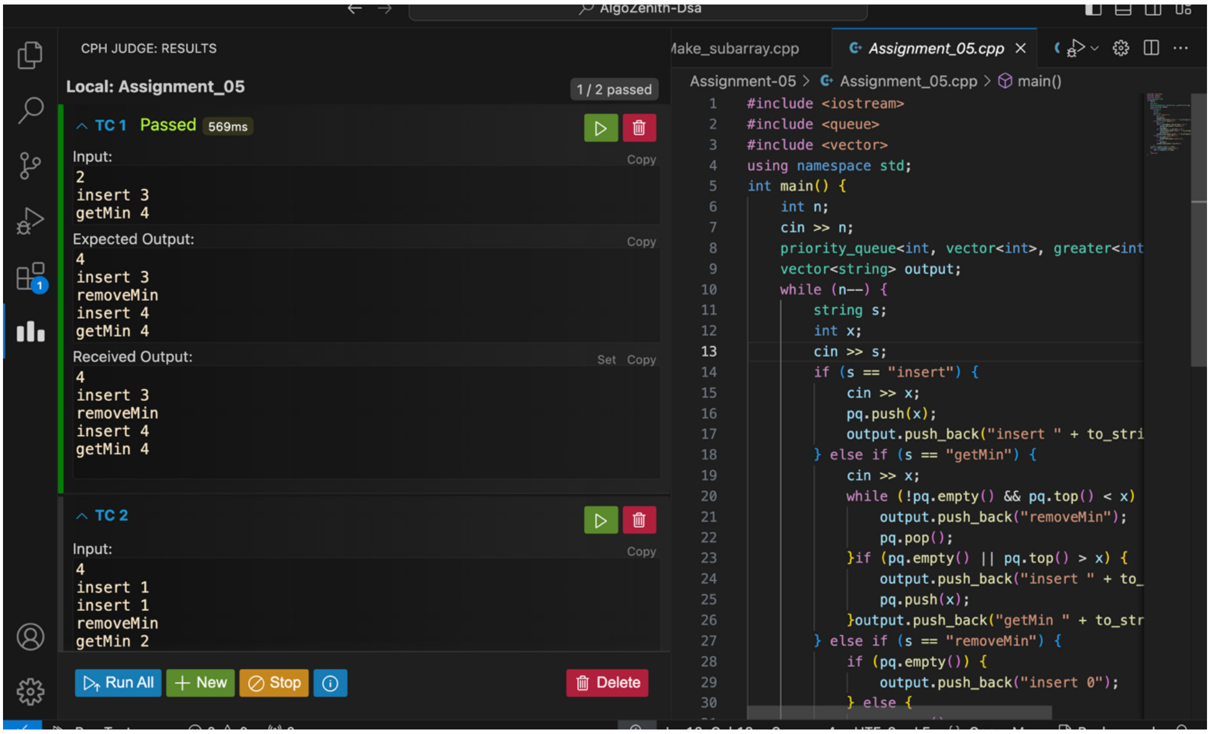


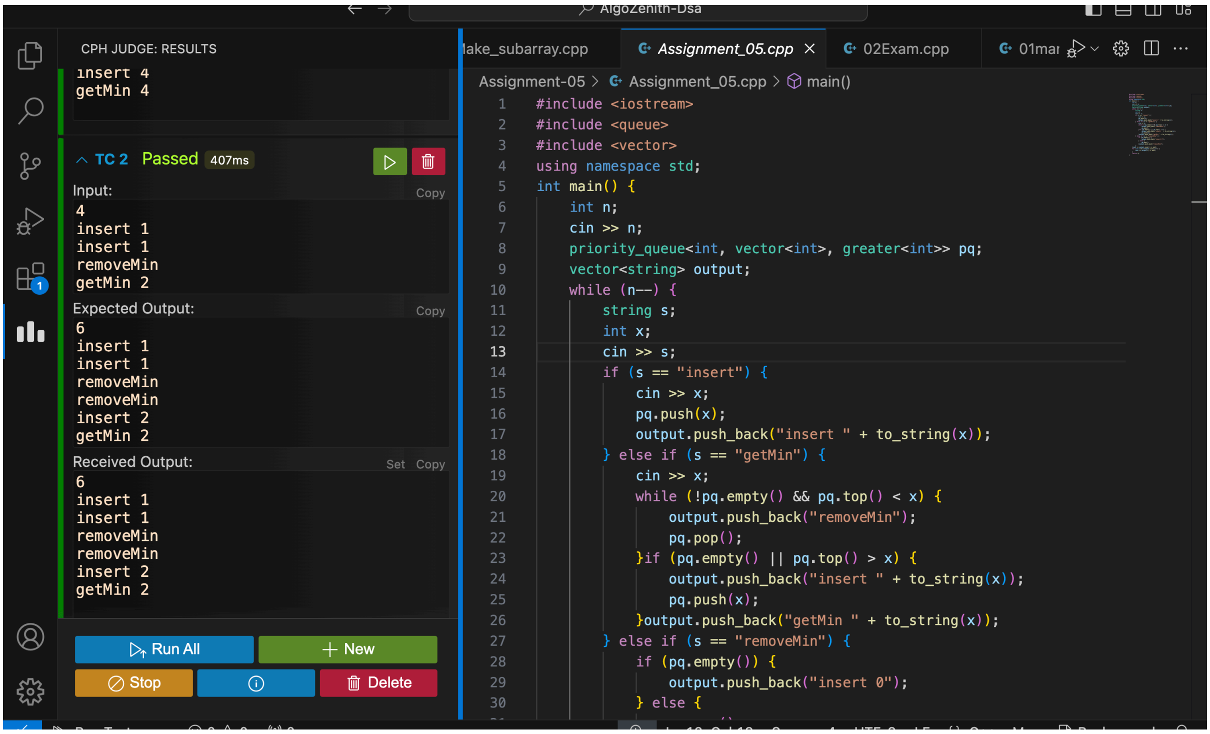
Assignment-05

1) Code:

```
#include <iostream>
#include <queue>
#include <vector>
using namespace std;
int main() {
    int n;
    cin >> n;
    priority_queue<int, vector<int>, greater<int>> pq;
    vector<string> output;
    while (n--) {
        string s;
        int x;
        cin >> s;
        if (s == "insert") {
            cin >> x;
            pq.push(x);
            output.push_back("insert " + to_string(x));
        } else if (s == "getMin") {
            cin >> x;
            while (!pq.empty() && pq.top() < x) {
                output.push_back("removeMin");
                pq.pop();
            }if (pq.empty() || pq.top() > x) {
                output.push_back("insert " + to_string(x));
                pq.push(x);
            }output.push_back("getMin " + to_string(x));
        } else if (s == "removeMin") {
            if (pq.empty()) {
                output.push_back("insert 0");
            } else {
                pq.pop();
            }output.push_back("removeMin");
        }
    }cout << output.size() << endl;
    for (int i=0;i<output.size();i++) {
        cout << output[i] << endl;
    }
    return 0;
}
```

1) Output:


CPH JUDGE: RESULTS
Local: Assignment_05
1 / 2 passed
TC 1 Passed 569ms
Input:
2
insert 3
getMin 4
Expected Output:
4
insert 3
removeMin
insert 4
getMin 4
Received Output:
4
insert 3
removeMin
insert 4
getMin 4
TC 2
Input:
4
insert 1
insert 1
removeMin
getMin 2
Run All
New
Stop
Delete
Assignment_05.cpp
#include <iostream>
#include <queue>
#include <vector>
using namespace std;
int main() {
int n;
cin >> n;
priority_queue<int, vector<int>, greater<int
vector<string> output;
while (n--) {
string s;
int x;
cin >> s;
if (s == "insert") {
cin >> x;
pq.push(x);
output.push_back("insert " + to_stri
} else if (s == "getMin") {
cin >> x;
while (!pq.empty() && pq.top() < x)
output.push_back("removeMin");
pq.pop();
}if (pq.empty() || pq.top() > x) {
output.push_back("insert " + to_
pq.push(x);
}output.push_back("getMin " + to_str
} else if (s == "removeMin") {
if (pq.empty()) {
output.push_back("insert 0");
} else {


CPH JUDGE: RESULTS
insert 4
getMin 4
TC 2 Passed 407ms
Input:
4
insert 1
insert 1
removeMin
getMin 2
Expected Output:
6
insert 1
insert 1
removeMin
removeMin
insert 2
getMin 2
Received Output:
6
insert 1
insert 1
removeMin
removeMin
insert 2
getMin 2
Run All
New
Stop
Delete
Assignment_05.cpp
#include <iostream>
#include <queue>
#include <vector>
using namespace std;
int main() {
int n;
cin >> n;
priority_queue<int, vector<int>, greater<int>> pq;
vector<string> output;
while (n--) {
string s;
int x;
cin >> s;
if (s == "insert") {
cin >> x;
pq.push(x);
output.push_back("insert " + to_string(x));
} else if (s == "getMin") {
cin >> x;
while (!pq.empty() && pq.top() < x) {
output.push_back("removeMin");
pq.pop();
}if (pq.empty() || pq.top() > x) {
output.push_back("insert " + to_string(x));
pq.push(x);
}output.push_back("getMin " + to_string(x));
} else if (s == "removeMin") {
if (pq.empty()) {
output.push_back("insert 0");
} else {